

Ethical Analysis: Submission 2

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Step 1: Draft 3 LLM Prompts

Prompt 1 → Descriptive

Prompt:

What are the ethical issues involved in Meta/Facebook using algorithms to decide which consumers receive housing advertisements? Explain some of the benefits and risks of using algorithms in advertising, and discuss how this issue could affect consumers, advertisers, Meta, regulators, civil rights organizations, and society.

Rationale:

I want the LLM to explain the ethical issues surrounding the use of algorithms in housing advertising. I also want to see if it discusses both the benefits and risks of the technology and considers how different stakeholders could be affected.

Prompt 2 → Comparative

Prompt:

How might Meta, advertisers, and consumers view the use of algorithms to deliver housing advertisements differently? Compare their perspectives on the benefits and ethical concerns, including advertising efficiency, personalization, fairness, and equal access to housing information.

Rationale:

I want to see how different stakeholders might have different views about the same ethical issue. I also want to see whether looking at the issues from different perspectives changes how the conflict between advertising efficiency and fairness is understood.

Prompt 3 → Hypothetical

Prompt:

Imagine Meta is creating a new algorithm to deliver housing advertisements. The algorithm is supposed to make advertising more efficient and relevant to consumers, but testing shows that some demographic groups are less likely to receive certain housing advertisements. What ethical concerns

could this create? Consider the perspectives of consumers, advertisers, Meta, regulators, civil rights organizations, and society.

Rationale:

I want to see how the LLM handles a situation where an algorithm has benefits but could also create unequal outcomes. I also want to see how the different stakeholders might view the situation and what ethical concerns the LLM identifies.

Step 2: Run Three Prompts through LLM and Analyze Responses

Response:

LLM Response — Prompt 1

The use of algorithms to decide who receives housing advertisements raises several ethical issues, especially around fairness, discrimination, transparency, and equal access to housing opportunities. Algorithms can make advertising more efficient by helping advertisers reach people who are more likely to be interested in a particular property. However, if an algorithm uses demographic information, user behavior, location, or other data in ways that result in certain groups being less likely to receive housing advertisements, it could create unequal access to housing information.

One major ethical concern is **algorithmic bias**. Algorithms learn from data and patterns, which means they can potentially reproduce existing biases or create unequal outcomes even when discrimination is not intentionally programmed into the system. In the case of housing advertisements, this could mean that some consumers are shown fewer opportunities or are excluded from seeing certain housing advertisements based on characteristics associated with protected groups.

Another concern is **transparency**. Consumers may not know why they are seeing certain advertisements or why they are not seeing others. Advertisers and platforms may also have difficulty explaining exactly how an algorithm makes decisions. This can make it harder to identify or address unfair outcomes.

There are also benefits to using algorithms in advertising. Algorithms can help advertisers reach relevant audiences, reduce advertising costs, personalize advertisements, and make advertising more efficient. Consumers may also benefit from receiving advertisements that are more relevant to their interests. For Meta, using algorithms can improve the effectiveness of its advertising platform and provide value to advertisers.

Different stakeholders are affected in different ways. **Consumers** could benefit from relevant advertisements but could also be harmed if biased algorithms limit their access to housing opportunities. **Advertisers** benefit from more targeted and efficient advertising but could face ethical, legal, and financial consequences if their advertising reaches people unfairly. **Meta** benefits from the efficiency and revenue generated by its advertising system but has a responsibility to address potential bias and discrimination on its platform. **Regulators and civil rights organizations** have an interest in protecting

consumers from discrimination and promoting equal access to housing opportunities. **Society** is affected because unequal access to housing information can contribute to broader inequalities.

Overall, the main ethical conflict is between the **efficiency and personalization provided by algorithmic advertising** and the need for **fairness, transparency, and equal access**. The use of algorithms is not necessarily unethical by itself, but the outcomes and decision-making processes need to be considered carefully, especially when the advertising involves important opportunities such as housing.

Analysis:

As for the response being clear and relevant, in reference to prompt 1, it was clear and directly answered the prompt. This response discussed both the **benefits and risks** of using algorithms in advertising. This also addressed all of the stakeholders I asked it to consider: consumers, advertisers, Meta, regulators, civil rights organizations, and society. This response seemed to be organized around the main ethical concerns, which made it easier to follow.

Next, for the analysis identifying ethical dimensions, it did identify certain ethical dimensions. First being, **algorithmic bias** as well as the possibility of unequal outcomes. **Discrimination** being another one, especially if certain groups happened to receive fewer housing opportunities. **Transparency** was another one, since consumers may not know why they see or do not see certain ads. **Fairness and equal access** was another ethical dimension, especially when it comes to housing information. The response also identified the conflict between **advertising efficiency/personalization and fairness**. Finally, this also recognized that algorithms can have benefits, such as making advertising more relevant and efficient.

The response did not leave out any major ethical issues, although it did not go very deeply into the **specific Meta housing-advertising case** from my research. This response mostly discussed the issue generally rather than connecting it to the specific events involving Meta. This response did not discuss the **legal/regulatory side** of the issue in much detail. This response presented algorithms as beneficial for efficiency and personalization, while focusing more heavily on the potential risks to fairness. Lastly, I found it interesting that the response said the use of algorithms is **not automatically unethical** and that the outcomes and decision-making process should be considered.

Response:

LLM Response — Prompt 2

Meta, advertisers, and consumers could view the use of algorithms to deliver housing advertisements differently because each stakeholder has different interests and concerns.

Meta may view algorithms as a way to make its advertising platform more efficient and useful for advertisers. Algorithms can help determine which advertisements are most relevant to different users and can improve the performance of advertising campaigns. However, Meta also has to consider ethical concerns such as fairness, discrimination, transparency, and equal access to housing information. If the

algorithm creates unequal outcomes for certain groups, Meta could face criticism or legal and regulatory concerns.

Advertisers may focus more on the benefits of reaching people who are likely to be interested in their housing advertisements. Algorithmic targeting can help advertisers use their advertising budgets more efficiently and potentially reach consumers who are more likely to respond to their ads. At the same time, advertisers may be concerned that an algorithm could unintentionally exclude certain groups from seeing their advertisements. This could create ethical concerns about fairness and equal access, especially when advertising housing opportunities.

Consumers may have a different perspective because they are the people receiving the advertisements. Personalization can be helpful because consumers may see housing opportunities that are relevant to their interests or location. However, consumers may also be concerned about what information is being used to determine which advertisements they see. If an algorithm causes certain consumers to receive fewer housing advertisements, they could have less access to housing opportunities than other groups.

The main difference between these perspectives is what each stakeholder stands to gain or lose. **Meta and advertisers may focus more on efficiency, personalization, and the effectiveness of advertising**, while **consumers may place more importance on fairness and equal access to information**. There can also be overlap because all three groups can benefit from an advertising system that is effective and relevant.

Overall, the ethical concern comes from balancing the benefits of algorithmic advertising with the potential for unequal outcomes. Algorithms can make housing advertising more efficient and personalized, but those benefits need to be considered alongside fairness, transparency, and equal access to housing information.

Analysis:

As for the prompt 2 response being clear and relevant, this response also was clear and directly answered this comparative prompt. This response compared the perspectives of **Meta, advertisers, and consumers** separately. The response also discussed the benefits and ethical concerns for each stakeholder. It also addressed the specific topics from this prompt: **efficiency, personalization, fairness, and equal access**. This response helped make it clear that each stakeholder may have different priorities based on what they could gain or could lose.

This prompt and response also covered some of the identifiable ethical dimensions. First being, **fairness**, as far as how housing advertisements are being delivered. Also, **discrimination** and the possibility of certain groups being excluded. **Transparency** as well, especially regarding how algorithms decide which ads people even see. Making sure there is **equal access** to housing information and opportunities is also a vivid one. **Privacy/data use** is also another ethical concern because consumers may be concerned about what information is used to determine the ads they receive. The response also discussed the conflict between **advertising efficiency/personalization and fairness**. Finally, the response also recognized that there can be **overlap between stakeholder interests**, since Meta, advertisers, and consumers can all benefit from relevant and effective advertising.

In conclusion, I would have to say that this response did not go into much detail about the **specific Meta housing-advertising case** from my research. The response stayed fairly general and focused on comparing the different stakeholder perspectives rather than specific actions Meta or advertisers have taken. This also did not discuss the **legal/regulatory consequences** in much detail. This focused on only three stakeholders, which matches the prompt, and therefore did not bring in **regulators, civil rights organizations, or society** like Prompt 1 did. One interesting point I noticed was that the response recognized that **Meta and advertisers may focus more on efficiency and effectiveness, while consumers may focus more on fairness and equal access**. Another interesting point made was that the response did not treat the stakeholders as completely opposed; it pointed out that **they can have overlapping interests**.

Response:

LLM Response — Prompt 3

If testing shows that some demographic groups are less likely to receive certain housing advertisements, several ethical concerns could arise. The main concerns would include **fairness, discrimination, equal access, transparency, and accountability**. Even if the algorithm was not intentionally designed to discriminate, unequal outcomes could still affect people's access to housing opportunities.

Consumers could be affected if certain groups receive fewer housing advertisements or are excluded from opportunities that other consumers see. This raises concerns about fairness and equal access to housing information. Consumers may also want to know why they are receiving certain advertisements and what information is being used to make those decisions.

Advertisers could benefit from a more efficient algorithm because it could help them reach consumers who are more likely to be interested in their housing advertisements. However, advertisers could also be concerned if the algorithm unintentionally excludes certain groups. They may have to balance the benefits of targeted advertising with the responsibility to make sure their advertisements are reaching audiences fairly.

Meta could benefit from creating a more efficient and personalized advertising system, but it would also have a responsibility to monitor the algorithm for unequal outcomes. If testing shows that certain groups are being affected differently, Meta may need to investigate why this is happening and consider changes to the algorithm.

Regulators may be concerned about whether the algorithm creates discriminatory outcomes or violates laws designed to protect people from discrimination in housing. They may also be interested in whether Meta has enough oversight and accountability for how its algorithms affect consumers.

Civil rights organizations could focus on whether certain demographic groups are being denied equal access to housing information. They may be concerned that algorithmic systems could reinforce existing inequalities, even if discrimination is not intentional.

Society could be affected more broadly if unequal access to housing advertisements contributes to existing social or economic inequalities. Housing is an important opportunity, so unequal access to information about available housing could have effects beyond advertising itself.

There are also benefits to using the algorithm. It could make advertising more efficient, reduce wasted advertising spending, and provide consumers with housing advertisements that are more relevant to their interests. However, these benefits have to be considered alongside the potential ethical concerns.

Overall, the main ethical issue is the conflict between **efficiency and personalization on one hand and fairness and equal access on the other**. The fact that unequal outcomes were identified during testing suggests that the algorithm should be examined before being widely used. Ethical decision-making would require considering not only how well the algorithm performs, but also how its outcomes affect different groups.

Analysis:

After analyzing the 3rd prompt, this response was also clear and directly answered the hypothetical situation in the prompt. This response considered all of the stakeholders I asked it to: **consumers, advertisers, Meta, regulators, civil rights organizations, and society**. This response discussed both the **benefits and ethical concerns** of creating the algorithm. This response stayed focused on what could happen if testing showed that some demographic groups were less likely to receive housing advertisements. This response clearly explained the conflict between **efficiency/personalization as well as fairness/equal access**.

To discuss the identifiable ethical dimensions in this prompt, first being: **Fairness**, because some groups could receive fewer housing advertisements. Next, again, **discrimination**, even if the algorithm was not intentionally designed to discriminate. As stated earlier, **equal access** as far as housing information and opportunities. **Transparency**, as earlier stated, including consumers wanting to know why they receive certain advertisements. **Accountability**, especially due to Meta's responsibility to monitor the algorithm. **Privacy/data use** is also an ethical concern because consumers may be concerned about what information is being used. Lastly, **existing inequalities** are another ethical concern because unequal access to housing information could contribute to broader social or economic inequalities. This response clearly explained the conflict between **efficiency/personalization and fairness/equal access**.

In conclusion, this third prompt response did not go very deeply into the **specific Meta housing-advertising case** from my research. This response did not provide many specific examples of what Meta could actually do to address the unequal outcomes beyond investigating the algorithm and considering changes. This response did not discuss the **specific DOJ legal case or settlement** from my research. This focused more on what *could* happen in the hypothetical situation rather than connecting it to what has actually happened with Meta. One interesting point made was that the response recognized that **discrimination does not necessarily have to be intentional for an algorithm to create unequal outcomes**. Another standout point was that it said the algorithm should be evaluated based on **not only how well it performs but also how its outcomes affect different groups**.

Step 3: Refine and Re-run 2 Prompts

Step 3 → Prompt 1 Redefined

Original Prompt 1:

What are the ethical issues involved in Meta/Facebook using algorithms to decide which consumers receive housing advertisements? Explain some of the benefits and risks of using algorithms in advertising, and discuss how this issue could affect consumers, advertisers, Meta, regulators, civil rights organizations, and society.

Refined: (Want to add more context from research and explicitly identify the stakeholders, without bias)

Refined Prompt 1:

Prompt 1:

Meta/Facebook has used algorithms to determine which consumers receive housing advertisements. Research into this issue has raised concerns about algorithmic bias, discrimination, fairness, transparency, and equal access to housing information. At the same time, algorithmic advertising can provide benefits such as efficiency, personalization, and more relevant advertising. Analyze the ethical issues involved in this situation and consider the perspectives of consumers, advertisers, Meta, regulators, civil rights organizations, and society. Explain the potential benefits and risks and discuss the conflict between advertising efficiency and fairness.

Neutral Version:

How should the ethical issues surrounding Meta/Facebook's use of algorithms to deliver housing advertisements be evaluated? Consider the potential benefits of algorithmic advertising, such as efficiency, personalization, and relevant advertising, as well as potential concerns related to algorithmic bias, discrimination, fairness, transparency, and equal access to housing information. Consider the perspectives of consumers, advertisers, Meta, regulators, civil rights organizations, and society. What factors should be considered when evaluating whether the use of algorithms produces fair and ethical outcomes?

Biased Version:

Why is Meta/Facebook's use of algorithms to deliver housing advertisements ethically problematic? Consider how algorithmic bias, discrimination, and unequal access to housing information can negatively affect consumers and society. Discuss the risks of allowing Meta to use these algorithms, including potential harm to protected groups, and explain why fairness and equal access should be prioritized over advertising efficiency and personalization.

Analysis Refinement:

Prompt 1 was chosen to be refined due to wanting to add more context from my research so the LLM had a better understanding of the specific ethical issue being discussed. I also wanted to clearly identify the stakeholders as well as include both the benefits and risks of algorithmic advertising. I chose to keep the prompt neutral so the LLM could analyze the issue without being pushed toward a specific conclusion.

Step 3 → Re-run and Compare LLM Responses

Prompt 1 → Neutral Version:

How should the ethical issues surrounding Meta/Facebook's use of algorithms to deliver housing advertisements be evaluated? Consider the potential benefits of algorithmic advertising, such as efficiency, personalization, and relevant advertising, as well as potential concerns related to algorithmic bias, discrimination, fairness, transparency, and equal access to housing information. Consider the perspectives of consumers, advertisers, Meta, regulators, civil rights organizations, and society. What factors should be considered when evaluating whether the use of algorithms produces fair and ethical outcomes?

LLM Response — Neutral Version

The ethical issues surrounding the algorithm should be evaluated by considering both the benefits of using algorithmic advertising and the potential risks of unequal outcomes. Algorithms can make housing advertising more efficient by helping advertisers reach consumers who may be interested in specific properties. Personalization can also make advertisements more relevant to consumers and help advertisers use their advertising budgets more effectively. However, these benefits need to be considered alongside concerns about fairness, discrimination, transparency, equal access, and accountability.

One important factor is whether the algorithm creates unequal outcomes for different demographic groups. If some groups are less likely to receive certain housing advertisements, they may have less access to information about housing opportunities. Even if the algorithm was not intentionally designed to discriminate, the data or decision-making processes used by the system could contribute to unequal outcomes. The extent and causes of these differences should therefore be investigated.

Transparency is another factor that should be considered. Consumers may not understand why they receive certain housing advertisements or why they do not receive others. Meta and advertisers should be able to explain how the algorithm makes decisions and should have processes for identifying and addressing potential bias. Accountability is also important because there should be clear responsibility for monitoring the algorithm and responding when unequal outcomes are identified.

Different stakeholders may view the issue differently. **Consumers** may be concerned about having fair and equal access to housing information. **Advertisers** may value efficiency and reaching relevant audiences but may also be concerned about unintentionally excluding certain groups. **Meta** may benefit from an efficient and effective advertising system while also having responsibilities related to fairness, transparency, and accountability. **Regulators** may focus on whether the algorithm creates discriminatory outcomes or conflicts with legal protections. **Civil rights organizations** may be particularly concerned

about equal access and whether algorithmic systems reinforce existing inequalities. **Society** may be concerned about the broader effects of unequal access to housing opportunities.

Overall, determining whether the algorithm produces fair and ethical outcomes would require considering more than just how efficiently it delivers advertisements. The benefits of efficiency, personalization, and relevance should be weighed alongside the effects on different demographic groups. Factors such as the size and consistency of unequal outcomes, the reasons those outcomes occur, transparency in the algorithm's decision-making, accountability for addressing problems, and equal access to housing information should all be considered. The goal should be to evaluate both the performance of the algorithm and its impact on the different groups affected by it.

Analysis → Neutral Version:

The neutral version provided a balanced response because it considered both the benefits and risks of algorithmic advertising. The response discussed efficiency and personalization while also addressing algorithmic bias, discrimination, fairness, transparency, and equal access. It also considered the perspectives of the different stakeholders rather than assuming that one stakeholder's interests were more important than another's.

The response was also more focused on evaluating the issue than automatically labeling the algorithm as ethical or unethical. Similar to the original response, it recognized that algorithms can provide benefits while still creating ethical concerns depending on their outcome. The response also emphasized that discrimination does not necessarily have to be intentional for an algorithm to produce unequal results.

Prompt 1 → Biased Version:

Imagine Meta is developing an algorithm to deliver housing advertisements, but testing shows that some demographic groups are less likely to receive certain housing advertisements. Why should this algorithm be considered ethically problematic? Discuss how the algorithm could contribute to discrimination, unequal access to housing information, and existing inequalities. Consider the potential harm to consumers and society and explain why Meta should prioritize fairness and equal access over the benefits of efficiency and personalization.

LLM Response — Biased Version

The algorithm should be considered ethically problematic because testing has shown that some demographic groups are less likely to receive certain housing advertisements. This could create unequal access to housing information and potentially contribute to discrimination, even if the algorithm was not intentionally designed to discriminate. When certain groups receive fewer housing advertisements, they may have fewer opportunities to learn about available housing than other groups.

Algorithmic systems can also contribute to existing inequalities because they may rely on data and patterns that reflect inequalities that already exist. If those patterns influence which consumers receive housing advertisements, the algorithm could reinforce disadvantages for certain groups. This is especially

concerning in housing because access to housing opportunities can have significant effects on individuals, families, and communities.

The potential harm to consumers includes having fewer opportunities to learn about housing that may be relevant to them. Consumers may also be unaware that an algorithm is influencing the advertisements they receive, which makes the lack of transparency an additional ethical concern. If consumers cannot understand or challenge how advertising decisions are made, it may be more difficult to identify and address unfair outcomes.

Meta should prioritize fairness and equal access because efficiency and personalization should not come at the expense of potentially excluding certain groups from housing information. Although targeted advertising can benefit advertisers and Meta by making advertising more efficient, those benefits do not eliminate the potential harm caused by unequal access. Housing is an important opportunity, so preventing discriminatory outcomes should be given significant consideration.

Meta should therefore take steps to identify why certain demographic groups are receiving fewer housing advertisements and make changes to reduce discriminatory or unequal outcomes. Transparency, monitoring, and accountability would also be important to ensure that the algorithm does not continue producing unfair results.

Overall, the algorithm is ethically problematic because its unequal outcomes could contribute to discrimination and limit equal access to housing information. The potential benefits of efficiency and personalization need to be considered, but fairness and equal access should be prioritized when an algorithm affects access to important opportunities such as housing.

Analysis → Biased Version:

The biased version produced a noticeable difference in tone because the wording of the prompt already assumed that Meta's algorithm was ethically problematic. Instead of asking the LLM to evaluate whether the algorithm creates an ethical problem, the prompt asked why it should be considered problematic and specifically emphasized discrimination, harm, and unequal access.

Because of this wording, the response focused much more heavily on the risks of the algorithm. Although it mentioned efficiency and personalization, these benefits were presented as secondary to fairness and equal access. The response also used stronger language about the potential harm to certain demographic groups and the responsibility of Meta to prioritize fairness.

This shows how a biased prompt can influence an LLM's reasoning. The response was not necessarily based on different information from the neutral version; instead, the framing of the question encourages the LLM to focus on one side of the ethical issue.

Comparison of Neutral and Biased Responses → Prompt 1:

The neutral and biased responses discussed many of the same ethical issues, including algorithmic bias, discrimination, fairness, transparency, and equal access. However, their reasoning was different because of how the prompts were written. The neutral version considered both the benefits and

risks and asked what factors should be considered when evaluating the algorithm. The biased version started from the assumption that the algorithm was ethically problematic, so the response focused more heavily on potential discrimination and harm. This comparison shows that prompt wording can influence not only the tone of an LLM response but also which ethical concerns receive the most attention.

Step 3 → Prompt 3 Redefined

Original Prompt 3:

Imagine Meta is creating a new algorithm to deliver housing advertisements. The algorithm is supposed to make advertising more efficient and relevant to consumers, but testing shows that some demographic groups are less likely to receive certain housing advertisements. What ethical concerns could this create? Consider the perspectives of consumers, advertisers, Meta, regulators, civil rights organizations, and society.

Refined: (Want to make hypothetical scenario more specific & focus on possible outcomes of algorithm)

Refined Prompt 3:

Prompt 3:

Imagine Meta is developing an algorithm to deliver housing advertisements. Research into Meta's housing advertisements has raised concerns about algorithmic bias, discrimination, fairness, transparency, and equal access to housing information. At the same time, algorithmic advertising can provide benefits such as efficiency, personalization, and more relevant advertising. If testing shows that some demographic groups are less likely to receive certain housing advertisements, analyze the ethical concerns this could create. Consider the perspectives of consumers, advertisers, Meta, regulators, civil rights organizations, and society. Discuss both the potential benefits and risks of the algorithm and the conflict between advertising efficiency and fairness.

Neutral Version:

Imagine Meta is developing an algorithm to deliver housing advertisements. If testing shows that some demographic groups are less likely to receive certain housing advertisements, how should the ethical issues surrounding the algorithm be evaluated? Consider the potential benefits of the algorithm, such as efficiency, personalization, and relevant advertising, as well as concerns related to fairness, discrimination, transparency, equal access, and accountability. Consider the perspectives of consumers, advertisers, Meta, regulators, civil rights organizations, and society. What factors should be considered when evaluating whether the algorithm produces fair and ethical outcomes?

Biased Version:

Imagine Meta is developing an algorithm to deliver housing advertisements, but testing shows that some demographic groups are less likely to receive certain housing advertisements. Why should this

algorithm be considered ethically problematic? Discuss how the algorithm could contribute to discrimination, unequal access to housing information, and existing inequalities. Consider the potential harm to consumers and society and explain why Meta should prioritize fairness and equal access over the benefits of efficiency and personalization.

Analysis Refinement:

Prompt 3 was chosen to be refined because I wanted to make the hypothetical situation more specific and realistic. I wanted the LLM to think about what could happen if testing showed that certain groups were receiving fewer housing advertisements. Also, I wanted the refined prompt to focus more on the possible consequences of the algorithm rather than only identifying the general ethical concerns.

Step 3 → Re-run and Compare LLM Responses

Prompt 3 → Neutral Version:

Imagine Meta is developing an algorithm to deliver housing advertisements. If testing shows that some demographic groups are less likely to receive certain housing advertisements, how should the ethical issues surrounding the algorithm be evaluated? Consider the potential benefits of the algorithm, such as efficiency, personalization, and relevant advertising, as well as concerns related to fairness, discrimination, transparency, equal access, and accountability. Consider the perspectives of consumers, advertisers, Meta, regulators, civil rights organizations, and society. What factors should be considered when evaluating whether the algorithm produces fair and ethical outcomes?

LLM Response — Neutral Version

The ethical issues surrounding the algorithm should be evaluated by considering both its potential benefits and its potential effects on different demographic groups. The algorithm could make housing advertising more efficient by helping advertisers reach consumers who may be interested in specific properties. Personalization could also make advertisements more relevant to consumers and allow advertisers to use their advertising budgets more effectively. However, these benefits should be considered alongside concerns about fairness, discrimination, transparency, equal access, and accountability.

A major factor to consider is whether the algorithm is producing unequal outcomes. If some demographic groups are less likely to receive certain housing advertisements, those groups may have less access to information about available housing opportunities. Even if the algorithm was not intentionally designed to discriminate, the data or patterns used by the algorithm could contribute to unequal outcomes. It would therefore be important to investigate why these differences are occurring and whether they are consistent across different groups.

Transparency is another important consideration. Consumers may not know why they are receiving certain advertisements or why they are not receiving others. Meta and advertisers should have ways to understand and monitor how the algorithm makes decisions. Accountability is also important because there should be clear responsibility for identifying potential problems and responding when unequal outcomes are discovered.

The different stakeholders may view the issue from different perspectives. **Consumers** may be concerned about fair and equal access to housing information. **Advertisers** may value efficiency and personalization but may also be concerned about unintentionally excluding certain groups. **Meta** may benefit from an effective advertising system while also having responsibilities related to fairness and accountability. **Regulators** may focus on whether the algorithm creates discriminatory outcomes or conflicts with legal protections. **Civil rights organizations** may focus on equal access and the possibility that algorithmic systems could reinforce existing inequalities. **Society** may be concerned about the broader effects of unequal access to housing opportunities.

Overall, evaluating whether the algorithm produces fair and ethical outcomes would require looking beyond its efficiency and effectiveness. Factors such as the extent of unequal outcomes, the reasons those outcomes occur, the transparency of the decision-making process, accountability for addressing problems, and equal access to housing information should all be considered. The potential benefits of efficiency, personalization, and relevant advertising should be weighed against the potential effects on different groups. This would allow the ethical evaluation to consider both the benefits of the technology and its consequences for the people affected by it.

Analysis → Neutral Version:

The neutral version considered the hypothetical situation without immediately assuming that the algorithm was unethical. It discussed the benefits of efficiency, personalization, and relevance while also recognizing the concerns created by unequal outcomes. The response also considered all of the stakeholders included in the prompt and discussed how their priorities could differ.

One important part of the response was that it recognized that unequal outcomes do not automatically explain why they occurred. This made the response more analytical because it suggested that Meta would need to investigate the cause of the difference before determining what changes should be made. The response therefore focused on evaluating the situation rather than automatically reaching a conclusion.

Prompt 3 → Biased Version:

Imagine Meta is developing an algorithm to deliver housing advertisements, but testing shows that some demographic groups are less likely to receive certain housing advertisements. Why should this algorithm be considered ethically problematic? Discuss how the algorithm could contribute to discrimination, unequal access to housing information, and existing inequalities. Consider the potential harm to consumers and society and explain why Meta should prioritize fairness and equal access over the benefits of efficiency and personalization.

LLM Response — Biased Version

The algorithm should be considered ethically problematic because testing has shown that some demographic groups are less likely to receive certain housing advertisements. This creates a risk of discrimination and unequal access to housing information, even if the algorithm was not intentionally

designed to discriminate. If some groups consistently receive fewer housing advertisements, they may have fewer opportunities to learn about available housing than other groups.

The algorithm could also contribute to existing inequalities. Algorithms often rely on data and patterns from past behavior, which can reflect inequalities that already exist. If these patterns influence which consumers receive housing advertisements, the algorithm could reinforce disadvantages for certain demographic groups instead of treating consumers equally.

The potential harm to consumers is significant because housing information can affect access to important opportunities. Consumers who receive fewer advertisements may not know about housing options that could otherwise be available to them. They may also be unaware that an algorithm is influencing the information they receive, creating additional concerns about transparency and accountability.

Meta should prioritize fairness and equal access over efficiency and personalization because a more efficient advertising system does not necessarily produce fair outcomes. While efficiency and personalization can benefit Meta, advertisers, and consumers, these benefits should not come at the expense of excluding certain groups from housing information. Housing is an important opportunity, so preventing discriminatory outcomes should receive significant consideration.

Meta should therefore investigate why certain demographic groups are less likely to receive specific housing advertisements and take steps to address unequal outcomes. Monitoring the algorithm, increasing transparency, and creating accountability for its effects could help reduce the risk of continued discrimination.

Overall, the algorithm is ethically problematic because its unequal outcomes could contribute to discrimination, limit equal access to housing information, and reinforce existing inequalities. Although algorithmic advertising can provide benefits, fairness and equal access should be prioritized when the technology affects access to important opportunities such as housing.

Analysis → Biased Version:

The biased version reached a more direct conclusion than the neutral version. The prompt asked why the algorithm should be considered ethically problematic and specifically instructed the LLM to explain why fairness and equal access should be prioritized over efficiency and personalization. As a result, the response focused primarily on discrimination, unequal access, and potential harm.

The response still acknowledged the benefits of efficiency and personalization, but it treated them as less important than fairness and equal access. This was different from the neutral version, which left more room to consider whether the unequal outcomes were caused by the algorithm and what additional information would be needed to evaluate the situation.

Comparison of Neutral and Biased Responses → Prompt 3:

The neutral and biased versions both identified discrimination, fairness, transparency, and equal access as important ethical concerns. However, the neutral version was more focused on evaluating the

hypothetical situation and determining why the unequal outcomes occurred. The biased version was more focused on explaining why the algorithm was ethically problematic and why fairness should take priority over efficiency.

The biased prompt therefore pushed the LLM toward a stronger conclusion. This demonstrates that changing only the wording and framing of a prompt can influence how an LLM weighs competing ethical considerations, even when the underlying situation and information remain similar.

Step 4 → Reflection:

Overall, refining and re-running the prompts showed me how much the way a question is written can affect an LLM's analysis of an ethical issue. Adding more context about algorithmic bias, discrimination, fairness, transparency, and equal access gave the LLM a better understanding of the specific issue and allowed it to provide a more focused analysis. Including the different stakeholders also showed how the same issue can be viewed differently depending on who is affected. Consumers and civil rights organizations may focus more on fairness and equal access, while advertisers and Meta may place more value on efficiency, personalization, and relevant advertising.

The comparison between the neutral and biased prompts showed me that the wording of a prompt can influence not only the tone of an LLM's response, but also which ethical concerns receive the most attention. The neutral prompts allowed the LLM to consider both the benefits and risks of algorithmic advertising without assuming a specific conclusion. The biased prompts, however, used wording that already suggested the algorithm was ethically problematic, which could lead the LLM to focus more heavily on discrimination and potential harm. This demonstrated that an LLM does not analyze an ethical issue completely independently of the prompt. The information, wording, and perspective included in the prompt can shape the reasoning and conclusions it provides.

In conclusion, this process showed me that creating an effective prompt is important when using an LLM to analyze an ethical issue. Providing enough context and including multiple stakeholder perspectives can lead to a more detailed and balanced analysis, while biased wording can influence the direction of the response. This also showed me why LLM responses should be evaluated critically rather than automatically accepted as objective or unbiased.